DNA barcoding

The idea of DNA barcoding was first raised in 2003 when a team at the University of Guelph, Canada, suggested that it would be possible to identify species by analyzing a common section of their DNA. Led by Dr. Paul Hebert, researchers chose a region in the gene known as cytochrome c oxidase 1 (“CO1”), made up of 648 base pairs. This region is quick to analyze, but the sequence is still long enough to differentiate between and within animal species. Different gene segments can be used for other forms of life.

The first part of the barcoding system involves cataloguing samples of known species. The DNA is extracted and organized into a sequence of base pairs, a process known as “sequencing.” The sequence is then stored in a computer database, so that when a DNA sample from an unknown species is sequenced and entered into the database, the computer will match it with existing records. The barcoding technique has proved useful for taxonomy, helping classify animals and plants.

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